

A-SMiLE: Affective Sparse Mixture-of-Experts Adapter with Multi-Task Learning for Spoken Dialogue Models

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Abstract

Large language models have made remarkable progress in generating coherent and contextually relevant dialogue. However, they still struggle to capture fine-grained paralinguistic cues, limiting emotional coherence and contextual appropriateness. To address this challenge, we propose an Affective Sparse Mixture-of-experts adapter with multi-task Learning (A-SMiLE) to enhance the spoken dialogue model’s capability for affective perception across multiple dimensions. Our approach integrates Valence, Arousal, and Dominance (VAD) modeling with response generation, thereby ensuring expressive and contextually aware interactions. Evaluations on DailyTalk and a hard-case benchmark demonstrate significant improvements over baselines in both emotion prediction and response generation. It shows the potential of cognitive-inspired affective modeling to enhance intelligent speech interactions on the spoken dialogue models.

Index Terms: Spoken Dialogue, Large Language Model, Representation Learning, Computational Paralinguistics, Mixture-of-Experts

1. Introduction

Large language models (LLMs) have revolutionized natural language processing, demonstrating remarkable capabilities in complex reasoning and dialogue generation [1]. However, spoken dialogue inherently conveys richer multimodal signals, integrating linguistic content with paralinguistic cues that critically shape communicative meaning [2].

Early LLM-based spoken dialogue systems [3] primarily process speech through textual representations, overlooking critical paralinguistic cues such as emotion, speaking style, and sentiment, leading to monotonous, unempathetic responses [4]. Despite recent efforts [2, 4, 5, 6] to integrate paralinguistics into LLM-based dialogue systems, challenges remain due to the complexity of human emotions and inherent constraints in data and models. Psychological research suggests that humans can experience approximately 34,000 distinct emotional states [7], yet existing approaches predominantly rely on predefined categorical labels or emotion representations from pretrained speech encoders. These methods are inherently restricted by the limited diversity of available emotional speech corpora, often leading to models that fail to capture fine-grained affective nuances, including subtle emotional shifts, sarcasm, and complex emotional states such as depression. As a result, these limitations can cause misalignment between model responses and conversational context, reducing the overall coherence and emotional appropriateness of generated dialogue. Furthermore, speech adapters, lightweight trainable networks that map speech representations into LLM-compatible embed-



Figure 1: An illustrative example of an empathetic LLM generating responses that adapt to the speaker’s emotional state, despite identical linguistic content in the input.

dings [8], are essential for integrating speech modality into large LLMs [9]. However, most existing adapters for spoken dialogue systems are primarily optimized for text-centric tasks, leading to task-specific overfitting [10, 11, 12] by prioritizing lexical alignment while disregarding essential paralinguistic cues. This bias limits the model’s ability to adapt to nuanced paralinguistic variations.

To bridge these gaps, we propose a novel framework that integrates multidimensional emotion perception into representation learning for LLM-based spoken dialogue systems, aligning with human cognitive and affective processing. Our approach is grounded in Russell’s circumplex model of affect [13], which characterizes emotions along Valence, Arousal, and Dominance (VAD), and the theory of affective neuroscience [14], which suggests that the brain exhibits functional specialization in processing different dimensions of emotional stimuli. By leveraging these principles, our model dynamically structures latent emotion representations to enable human-like affective perception, ensuring both emotional coherence and adaptability in response generation.

To achieve this, we employ a Cognitive-Inspired Affective Sparse Mixture-of-Experts (S-MoE) [15] adapter architecture combined with a multi-task learning approach, jointly modeling emotion representation learning and dialogue generation. Our proposed S-MoE architecture enables fine-grained affective control in spoken dialogue by dynamically activating specialized experts for different emotional dimensions (e.g., valence, arousal), ensuring responses that are both emotionally and contextually coherent. Specifically, our contributions are summarized as follows:

- Drawing on cognitive theories of affect processing, we leverage S-MoE to dynamically structure representation learning, enabling human-like affective perception and enhancing the model’s ability to capture fine-grained emotional nuances.
- We develop an Emotionally Adaptive Multi-Task Learn-

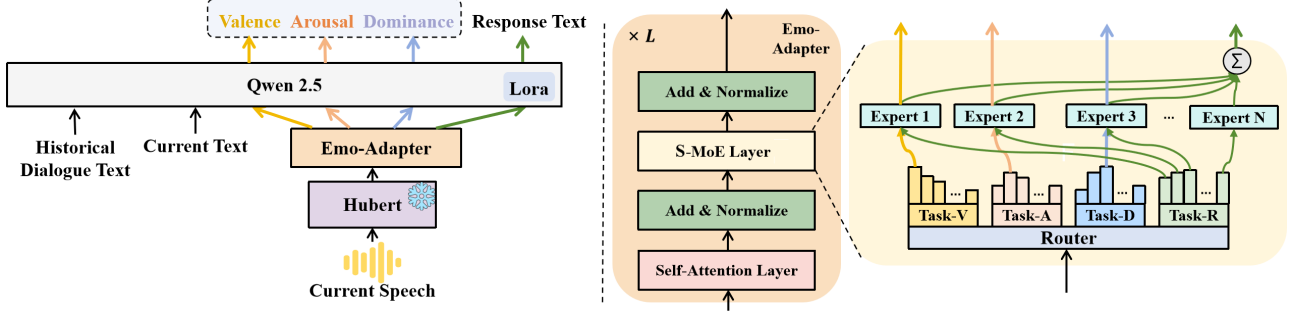


Figure 2: The overview of affective sparse mixture-of-experts adapter with multi-task learning (A-SMILE) for spoken dialogue models.

ing (EAML) framework, which dynamically integrates emotion perception with response generation, ensuring responses adaptively align with the affective context in real-time interactions.

- We develop a hard-case emotional dataset for emotion-sensitive spoken dialogue, incorporating sarcasm and complex emotional states. Experimental results on this dataset and DailyTalk [16] show that our model consistently surpasses baseline methods, demonstrating its effectiveness in nuanced emotional comprehension and its potential to improve human-machine interaction.

2. Proposed Method

2.1. Overall Architecture

As depicted in Figure 2, our framework integrates a speech encoder with an S-MoE adapter to enable human-like affective representation learning. The speech encoder processes input speech, and the resulting representations are refined by the MoE-based adapter, which dynamically selects specialized experts based on affective cues. This enhances the model’s ability to understand nuanced emotions and generate adaptive responses. The transformed features are subsequently aligned with the input space of an LLM. Within a multi-task learning paradigm—comprising VAD prediction and response generation—the LLM jointly optimizes emotion modeling and dialogue response generation, ensuring that responses are both semantically appropriate and emotionally attuned.

2.2. Cognitive-Inspired Affective S-MoE Adapter

Distinct brain regions process different affective dimensions: Valence is primarily regulated by the amygdala and prefrontal cortex [19], Arousal is modulated by the ventromedial prefrontal cortex (vmPFC) and limbic structures [20], and Dominance is associated with the anterior insula [21]. The brain dynamically engages specialized neural circuits to regulate emotional experiences, supporting the idea that affective processing is distributed across functionally distinct neural regions.

Inspired by this cognitive principle, we propose an Affective S-MoE Adapter to enable dynamic and efficient emotion representation learning within a multi-task learning framework. The architecture leverages task-aware expert activation to learn specialized representations for valence, arousal, and dominance, while integrating these dimensions to generate emotionally coherent responses.

The adapter employs a Transformer-based architecture with L repeated blocks, where each block integrates a S-MoE layer for dynamic routing. Given an input sequence $\mathbf{x}_s$, the gating

function computes routing logits as:

$$l_t(\mathbf{x}_s) = \mathbf{x}_s \cdot \mathbf{W}_{g,t}^T \quad (1)$$

Here, $\mathbf{W}_{g,t} \in \mathbb{R}^{N \times H}$ represents trainable task-specific weight matrices, where N is the number of experts and H is the hidden state dimension. The logits are normalized into probabilities over N experts using a softmax function:

$$G_t(\mathbf{x}_s)_i = \frac{\exp(l_t(\mathbf{x}_s)_i)}{\sum_{j=1}^N \exp(l_t(\mathbf{x}_s)_j)} \quad (2)$$

For each task t , the top- k experts are selected based on routing probabilities. This ensures efficient specialization and disentangled feature learning: For V, A, and D prediction, only the top-1 expert is activated per dimension. For response generation, the model aggregates outputs from the top-4 experts, enabling cross-dimensional synthesis for emotionally coherent responses. The output of the S-MoE layer is computed as:

$$h(\mathbf{x}_s) = \sum_{i \in \text{top-}k} G_t(\mathbf{x}_s)_i E_i(\mathbf{x}_s) \quad (3)$$

where $E_i(\mathbf{x}_s)$ denotes the output of expert i . This task-aware routing mechanism ensures efficient utilization of model capacity while enhancing emotion-sensitive representation learning.

2.3. Two-Stage Multi-Task Learning Framework

2.3.1. Stage 1: Alignment

In the first stage, training focuses on aligning the speech embedding with the LLM’s input space. To achieve this, the LLM remains frozen while only the speech adapter ϕ is optimized. During this phase, Cross-Entropy (CE) loss is applied for this speech adapter optimization.

2.3.2. Stage 2: EAML

Once the LLM successfully processes speech embeddings, the second stage fine-tunes both the LoRA [22] adapter θ and the speech adapter ϕ , enabling emotionally adaptive learning for spoken dialogue.

Our framework employs a multi-task learning paradigm, integrating emotion perception tasks (VAD prediction) as auxiliary tasks while treating response generation as the primary task. We define the joint loss function as:

$$\mathcal{L} = \lambda_e \mathcal{L}_e + \lambda_r \mathcal{L}_r \quad (4)$$

where $\mathcal{L}_e$ is the emotion prediction loss (Mean Squared Error, MSE loss), and $\mathcal{L}_r$ is the response generation loss. λ_e and λ_r control the balance between emotion modeling and response

Table 1: Evaluation Results on the hard-case emotional dataset

Method	V,A,D Prediction			Response Generation		
	CCC _V ↑	CCC _A ↑	CCC _D ↑	METEOR ↑	ROUGE-L ↑	GPT4o-1120 ↑
USDM[5]	0.35	0.31	0.23	8.6	10.3	5.5
SpeechGPT[17]	0.33	0.30	0.21	7.9	9.5	5.1
Qwen-Audio[18]	0.42	0.34	0.28	8.1	10.0	5.2
Qwen2.5-7B + ours	0.48	0.39	0.43	8.9	10.5	6.1

Table 2: Evaluation Results on DailyTalk Test Set.

Backbone	Method	V,A,D Prediction			Response Generation		
		CCC _V ↑	CCC _A ↑	CCC _D ↑	METEOR ↑	ROUGE-L ↑	GPT4o-1120 ↑
Qwen2.5-0.5B	Text only	0.41	0.32	0.30	7.1	9.0	5.7
	Cascaded	-	-	-	7.4	9.3	6.1
	Ours	0.54	0.44	0.39	7.9	9.7	6.6
Qwen2.5-7B	Text only	0.46	0.37	0.31	8.5	10.2	6.0
	Cascaded	-	-	-	8.9	10.7	6.5
	Ours	0.63	0.53	0.49	9.7	11.8	7.1

generation. This joint optimization strategy enables the model to align emotion understanding with response generation, ensuring emotionally coherent and contextually adaptive outputs.

3. Experiments

3.1. Dataset

This study adopts a two-stage training approach with distinct datasets. Stage 1 trains the model on a 10,000-hour corpus comprising GigaSpeech [23], CommonVoice [24], and LibriSpeech [25] to align speech embeddings with the LLM’s input space. Stage 2 fine-tunes the model on DailyTalk, a multi-turn dialogue dataset. DailyTalk consists of 2,541 scripted dialogues between two speakers (one male, one female), totaling 20 hours of speech. In this dataset, we extract VAD labels using a pre-trained dimensional speech emotion recognition model [26].

For evaluation, we use two different test sets to assess the performance of both our model and baseline systems in spoken dialogue tasks. One is the DailyTalk test set, including 300 DailyTalk dialogues. Another is a hard-case emotional dataset comprising two challenging categories: (1) **Low-Valence, Low-Dominance Utterances**, which often correspond to emotionally complex states such as depression, selected from IEMO-CAP [27], and (2) **Sarcasm**, where meaning is highly context-dependent, making conventional models struggle due to their reliance on lexical cues, sourced from MUsTARD [28]. The dataset consists of a total of **0.8 hours** of speech.

3.2. Evaluation Metrics

To assess the model’s ability to comprehend both semantic content and paralinguistic emotional cues in speech while generating high-quality, empathetic responses, we employ a modified system prompt designed to emphasize empathy in dialogue generation. Evaluation is conducted using the GPT-4o-2024-11-20 model, which independently scores responses from different systems based on empathy on a 0-10 scale.

For response text generation, we use ROUGE-L [29] and METEOR [30] to measure content similarity with ground-truth responses. To evaluate VAD prediction, we adopt the Concordance Correlation Coefficient (CCC) score [31] as a performance metric.

3.3. Implementation Details

Our framework integrates Qwen2.5 (0.5B/7B) as the language model with a HuBERT Large speech encoder. Context processing retains a 4-turn history window. Optimization is performed using the Adam optimizer with a batch size of 64 and learning rates of 1e-3 for Stage 1 and 1e-4 for Stage 2. We apply LoRA parameterization ($r=8$) exclusively to the LLM components to enable efficient adaptation [32].

3.4. Baseline Approaches

We evaluate our approach against two baseline configurations using Qwen2.5 0.5B and 7B LLM as the backbone, namely Text-LLM (Text-Only) and Text-LLM (Cascaded).

Text-LLM (Text-Only) involves fine-tuning a text-based LLM on DailyTalk without incorporating paralinguistic information. It serves as a reference to assess the model’s performance when relying solely on lexical content without access to prosodic or emotional cues. Text-LLM (Cascaded) incorporates VAD labels as additional textual input for response generation to enable the model to predict response styles and text better.

3.5. Evaluation on hard-case emotional dataset

Table 1 presents the results for **VAD prediction** and **response generation**. Our proposed framework significantly outperforms existing paralinguistics-enhanced LLMs across all emotion dimensions, achieving $CCC_V = 0.48$, $CCC_A = 0.39$, and $CCC_D = 0.43$, demonstrating enhanced sensitivity to subtle emotional nuances.

For response generation, our model attains superior performance, surpassing **Qwen-Audio** by $\Delta CCC_V = 0.06$, $\Delta CCC_A = 0.05$, and $\Delta CCC_D = 0.15$. These improvements highlight the effectiveness of our framework in enhancing **affective alignment** and **response expressiveness**.

3.6. Advancing Affective Understanding in Spoken Dialogue

In Table 2, our model consistently outperforms Text-LLM (Text-Only) baselines, underscoring the fundamental limitation of lexically driven dialogue systems—their inability to capture

Table 3: A qualitative example where the model generates Speaker B’s response.

Dialogue Context	B: This is one of my favorite places to kick back after a quest. A: Wow, cool tiger. B: Yeah I have him since level 10. Anyway, if you have your own game character, we could hang out, maybe go on a quest.
Current Utterance	A: Sounds interesting (spoken in a worried and hesitant tone).
Response Generation	Text-LLM (Cascaded): Definitely! Let’s find a day when we can go together. SpeechGPT: That’s the spirit! Let me know if you want any tips or company on your runs! USDM: Oh, that sounds fun! You should definitely create a character and join us. Ours: No pressure, take your time to think it over. Ground truth: So..you will think about it?

Table 4: Ablation Study Results.

Architecture	Method	METEOR $\uparrow$	ROUGE-L $\uparrow$	GPT-4o $\uparrow$
Conv-Based Adapter[33]	-	7.3	9.3	5.8
Q-former [34]	-	7.4	9.5	5.7
Transformer	-	7.4	9.4	5.6
	+S-MoE	8.2	10.5	5.8
	+EAML	8.7	10.8	6.3
	+S-MoE&EAML	8.9	11.1	6.4
6-Layer Transformer	-	7.6	9.7	6.1
	+S-MoE	8.7	10.7	6.3
	+EAML	9.1	11.3	6.9
	+S-MoE&EAML	9.7	11.8	7.1

emotional depth beyond explicit textual cues, often leading to effectively misaligned responses. In contrast, our approach dynamically integrates speech-derived VAD features with response generation, enabling emotionally coherent and contextually appropriate responses. For response generation, the Text-LLM (Cascaded) approach, which appends VAD labels as auxiliary text, provides minor improvements over Text-Only approach but remains limited by its reliance on explicit emotion tags rather than learned affective representations. This highlights the advantage of direct paralinguistic modeling, which allows for adaptive and expressive dialogue generation that better aligns with the speaker’s affective state.

3.7. Qualitative Example

Table 3 presents a qualitative comparison of model responses to a hesitant speaker. Text-LLM (Cascaded), SpeechGPT, and USDM generate responses with a more assertive and encouraging tone, failing to align with the speaker’s hesitant emotional state. In contrast, our model effectively captures the speaker’s worry and hesitation, producing a response that acknowledges uncertainty and provides reassurance. This demonstrates that our method dynamically integrates paralinguistic cues, enabling a more emotionally aligned and contextually appropriate response.

3.8. Ablation Study

To evaluate the contributions of EAML and Cognitive-Inspired Affective S-MoE, we conduct an ablation study across multiple speech adapter architectures, including the Conv-Based Adapter, Q-Former, and Transformer encoders with 4-layer and 6-layer configurations. The objective is to analyze the independent and joint effects of S-MoE and EAML on emotion perception and response generation.

As shown in Table 4, removing either S-MoE or EAML results in a notable performance drop, underscoring their individual contributions. Without S-MoE, the model exhibits limited

affective sensitivity, leading to weaker response coherence and reduced emotional expressiveness. Similarly, removing EAML diminishes the model’s ability to adaptively leverage emotional cues, making its responses less empathetic. In contrast, incorporating both S-MoE and EAML achieves the highest performance across emotion classification accuracy, response coherence, and stylistic alignment, demonstrating their complementary roles. The results confirm that expert routing and multi-task learning jointly enhance spoken dialogue modeling by effectively capturing and utilizing paralinguistic information.

These findings highlight the significance of integrating cognitively guided expert activation with emotionally adaptive training to improve expressiveness and contextual alignment in spoken dialogue systems.

4. Conclusion

Inspired by the compartmentalized affective processing of the human brain, we introduce an emotion-enhanced spoken dialogue model, A-SMiLE, which applies an affective sparse MoE adapter. The proposed approach integrates emotionally adaptive multi-task learning and a cognitive-guided S-MoE for LLMs. Our approach facilitates fine-grained emotion perception and adaptive response generation by dynamically selecting specialized experts and optimizing emotion-aware representations. This advancement emphasizes the potential of leveraging cognitive principles to improve the emotional intelligence of spoken dialogue systems, paving the way for more nuanced and contextually aware speech interactions. Evaluations on DailyTalk and a hard-case emotional dataset demonstrate that our model surpasses baselines in capturing subtle affective nuances and producing contextually appropriate, empathetic responses. These observations underscore the critical role of cognitive-guided affective modeling in spoken dialogue systems and its transformative potential to redefine human-machine interactions, bridging the gap between artificial intelligence and emotionally intelligent communication.

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